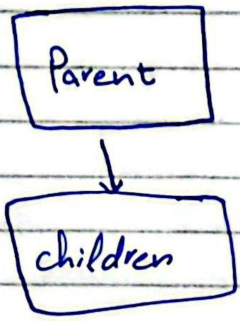
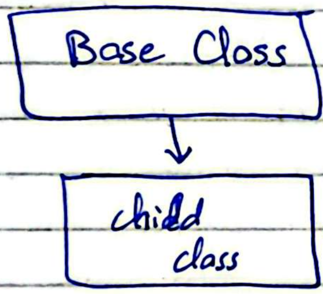
OOP

Inheritance:



like in real life when parents pass things to childrens same in case of class they can pass variables methods to childrens.



→ This can use variables and methods of Base class.

In Java we do this using extends Key words.

Q why we use inheritance?

Code reuseability:
Instead of writing same code in multiple classes we make a parent class having common properties then inherit it.

real life example is that like all user type of a software have age

So instead of user subclasses like child, adult, elder etc having an age variable we make a parent class then inherit that in child (sub classes).

-> "Super" Keyword:

it is used to call the parent class constructor and initialize values in parent class.
It means directly above it.

Imp point

-> Private Keyword in parent class

The variables declared as private in parent class or methods could not be inherited.

private bounds it to that specific file can not be used in another file.

Child class can access properties of parent class if they are not private but parent class cannot access the properties of child class.

UPCASTING AND DOWNCASTING:

if we do this

Parent P = new Child();

→ This is called upcasting
Now we can access parent class variables and methods but not child class because it is the reference that determind what we can access not the object

Parent P = new Child();

This will determine what we can access not child

Similarly

~~Child c = new Parent();~~

this is not possible because down casting is not allowed like that.

It is done as

~~Parent P = new C~~

~~Child~~

Parent P = new Child();

Child c = (Child) P;

-> Now I can access ~~dog~~ child as well as parent variables and methods.

-> In class Java every thing is inherited from object class.

Q: what If I have a same name variable in parent class as well as child class

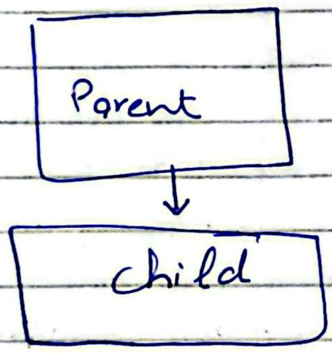
then for parent accessing we do super.name and for child we do this.name.

→ Super always called first in constructor because parent class is initialized first.

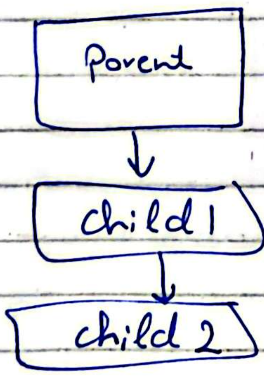
Constructors are not inherited in Java.

Types of Inheritance:

Single Inheritance:



Multiple Inheritance (Multilevel Inheritance)

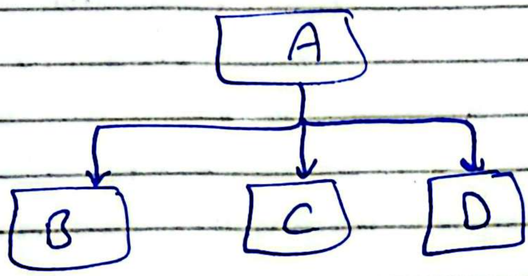


Java support ~~Multip~~ ~~Multip~~ Multilevel inheritance but not Multiple inheritance like

Class A extends B, C { }
we can do this with interfaces.

Hierarchial Inheritance:

One class is inherited by many classes



Hybrid Inheritance:

It is combination of single and multiple inheritance
So in Jove we know there is no multiple inheritance so no hybrid inheritance as well.